

Probing quantum coherence in a biological system by means of DNA amplification

Erhard Bieberich *

Institute of Molecular Medicine and Genetics, Medical College of Georgia, 1120 15th Street CB-2803, Augusta, GA 30912, USA

Received 25 March 2000; received in revised form 12 June 2000; accepted 20 June 2000

Abstract

As a result of rapid decoherence, quantum effects in biological systems are usually confined to single electron or hydrogen delocalizations. In principle, **molecular interactions at high temperatures can be guided by quantum coherence if embedded in a dynamics preventing decoherence.** This was experimentally investigated by analyzing the thermodynamics, kinetics, and quantum mechanics of the primer/template duplex formation during DNA amplification by polymerase chain reaction. The structures of the two oligonucleotide primers used for amplification of a cDNA template were derived either from a repetitive motif or a fractal distribution of nucleotide residues. Contrary to the computer-based calculation of the primer melting temperatures (T_m) that predicted a higher T_m for the non-fractal primer due to nearest-neighbor effects, it was found that the T_m of the non-fractal primer was actually 2°C lower than that of its fractal counterpart. A thermodynamic analysis of the amplification reaction indicated that the primer annealing process followed Bose–Einstein instead of Boltzmann statistics, with an additional binding potential of $\mu = 500$ J/mol or 10^{-21} J/molecule due to a superposition of binding states within the primer/template duplex. The temporal evolution of the Bose–Einstein state was determined by enzyme kinetic analysis of the association of the primer/template duplex to Taq polymerase. Assuming that collision with the enzyme interrupted the superposition, it was found that the Bose–Einstein state lasted for $t_{\text{dec}} = 0.7 \times 10^{-12}$ s, corresponding to the energy dispersion (ΔE) of quantum coherent states ($\mu = \Delta E \geq h/t_{\text{dec}}$). A quantum mechanical analysis revealed that the coherent state was stabilized by almost vanishing separation energies between distinct binding states during a temperature-driven shifting of the two DNA strands in the primer/template duplex. The additional binding potential is suggested to arise from a short-lived electron tunneling as the result of overlapping orbitals along the axis of the primer/template duplex. This effect was unique to the fractal primer due to the number of binding states that remained almost constant, irrespective of the size of shifting. It is suggested that fractal structures found in proteins or other macromolecules may facilitate a short-lived quantum coherent superposition of binding states. This may stabilize molecular complexes for rapid sorting of correct-from-false binding, e.g. during folding or association of macromolecules. The experimental model described in this paper provides a low-cost tool for simulating and probing quantum coherence in a biological system. © 2000 Elsevier Science Ireland Ltd. All rights reserved.

Keywords: Quantum mechanics; Quantum biology; Polymerase chain reaction; Enzyme kinetics; Fractal; Thermodynamics

* Tel.: +1-706-7219113; fax: +1-706-7218727.

E-mail address: ebieberich@mail.mcg.edu (E. Bieberich).

1. Introduction

The use of quantum computation for implementation of a fast search algorithm raises the question whether the quantum coherent superposition of different binding states can be exploited for molecular recognition in biological systems (Monroe et al., 1996; Ahn et al., 2000). The classical model describes the association/dissociation of molecular binding partners as a single collision, the association of which is energy cost intensive and diffusion-rate limited. In various biological systems, e.g. complementary base pairing in DNA, formation of a substrate–enzyme complex, or protein folding, molecular recognition is based on the simultaneous binding of several molecular sites resulting from a non-covalent interaction of distinct electron orbitals. A superposition of several possible binding states, e.g. by overlapping electron orbitals, could rapidly trace out the optimal complex conformation and sort correct-from-false binding. The direct experimental observation of a quantum coherent superposition in a biological system, however, is only reported in very rare cases such as single electron or hydrogen dispersion (Cha et al., 1989; Kohen et al., 1999; Ringe and Petsko, 1999). Most recently, it has been found that hydrogen tunneling due to a quantum superposition of two localized binding states of a hydrogen atom occurs independent of temperature between 30 and 65°C (Kohen et al., 1999). The tunneling effect was ‘catalyzed’ by the particular enzyme structure of thermophilic alcohol dehydrogenase. The experimental investigation of a superposition consisting of more than one or two electrons or atoms in a mesoscopic system is difficult due to rapid decoherence of the quantum coherent state at high temperatures. Conservative estimations predict decoherence times of 10^{-13} to 10^{-20} s as the life-time for a superposition in a biological system (Tegmark, 2000; Seife, 2000). Even if these states exist, they are indistinguishable from classical association/dissociation kinetics. In order to investigate a quantum coherent state of a biological system in vitro, it is necessary to stabilize a superposition of several binding states between two molecules. Furthermore, it must be possible

to experimentally determine the effect of the superposition for tracing out true/false binding states by quantization of the correct complex over time. Finally, the thermodynamics of the reaction must reveal a deviation from classical Boltzmann statistics in order to indicate a quantum coherent state during complex formation (Hill, 1986). Note that it is not intended to suggest a superposition of molecules themselves. However, a delocalization of binding electrons could contribute to an additional stabilization of a complex by forming overlapping orbitals in a quantum coherent state. This may be used for molecular recognition by ‘survival’ of most the stable state.

In this study, polymerase chain reaction (PCR) amplification of a template DNA with oligonucleotide primers containing a particular sequence periodicity was used to analyze the effect of the primer structure on the DNA duplex formation. The PCR approach was preferred over primer melting analysis due to its high sensitivity as the result of the amplification of short-lived effects arising from a potential binding state superposition. The sequence periodicity chosen for the oligonucleotide design was expected to facilitate the stabilization of superimposed binding states in the DNA duplex. As shown in Fig. 1, the oligonucleotide sequence was introduced at both ends of a template DNA by amplification with the primer combination R1a/R1s or F1a/F1s. These primers were composed of a gene specific sequence at the 3′-end and a sequence containing a distinct periodicity at the 5′-end. The two amplification products were then used as templates for a second round of PCR with the primers R2 and F2, which contained only the periodic part of the sequence (Fig. 1). As shown in Fig. 2, these two primers differed from each other in the periodicity of guanosine (g) residues interspersed in a string of adenosine (a) residues. Primer R2 contained alternating aa and gg pairs, whereas the distance of g residues in primer F2 was chosen on the basis of a golden means distribution (Peitgen et al., 1992). The ratio of the distances between g residues was nearly scale-invariant to the distance itself. The scale-invariant distribution of g residues conferred a self-similarity that is typical for fractal structures (Peitgen et al., 1992; Barnsley, 1993). Sup-

posing that mismatches can arise between the primer and template by shifting the two sequences along each other, the fractal structure will still match with a certain number of residues at each scale of shifting. It was expected that this topology rendered the F2/template duplex more stable than a duplex with R2. Note that the two primers differed only by a permutation of the fifth and sixth residue (Fig. 2). This permutation, despite conferring a self-similar structure to F2, resulted in a nearest-neighbor effect due to an additional gg pair in R2 (Aboul-ela et al., 1985; Breslauer et al., 1996; Rychlik et al., 1990; Rychlik, 1995). Thus, in contrast to the effects expected from a fractal topology, conventional programs for PCR primer design predicted a more stable duplex for R2 (Rychlik, 1995). The predicted melting temperature (T_m) was calculated to be 39.9°C for R2 and 39.1°C for F2. In order to evaluate these two contradictory predictions, PCR amplification was performed with a template of 500 base pairs (bp) at various annealing temperatures and the stability of the primer/template analyzed by determination of the amount of amplified DNA. The amplification reaction was performed with different concentrations of Taq polymerase in order to control the association kinetics of the enzyme

with the primer/template duplex. The thermodynamics and enzyme kinetics of the amplification reaction were analyzed in order to investigate Boltzmann versus Bose–Einstein statistics and classical versus quantum coherence in a biological model system.

2. Materials and methods

2.1. Materials

Platinum Taq polymerase, buffer, and deoxynucleotides were from Gibo BRL (Gaithersburg, MD). Oligonucleotide primers were synthesized by an in-house facility, and their purity analyzed by polyacrylamide gel electrophoresis and high-performance liquid chromatography before use. The PCR amplification was performed with a Crocodile III thermocycler (Applied Biosystems, Gaithersburg, MD). The sialyltransferase II (ST-II) specific template DNA was cloned from a library of mouse cDNA as described previously (Bieberich et al., 2000). All other reagents were of analytical grade or freshly redistilled before use. The sequence of the primers used was:

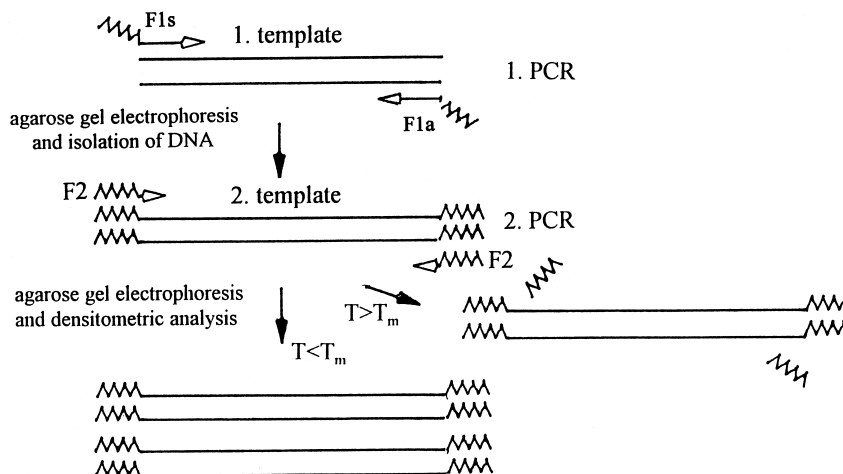


Fig. 1. Principle of PCR amplification. A cDNA specific for an enzyme investigated in our group was used as template with the primer combinations F1s(forward)/F1a(reverse) or R1s(forward)/R1a(reverse) (F = fractal, R = non-fractal) for the generation of the 2. template endowed with fractal or non-fractal sequence extensions. These extensions were used for annealing to F2 or R2 in the second round of PCR. The templates were purified by agarose gelelectrophoresis and the sequence determined by Sanger-dideoxy DNA sequencing.

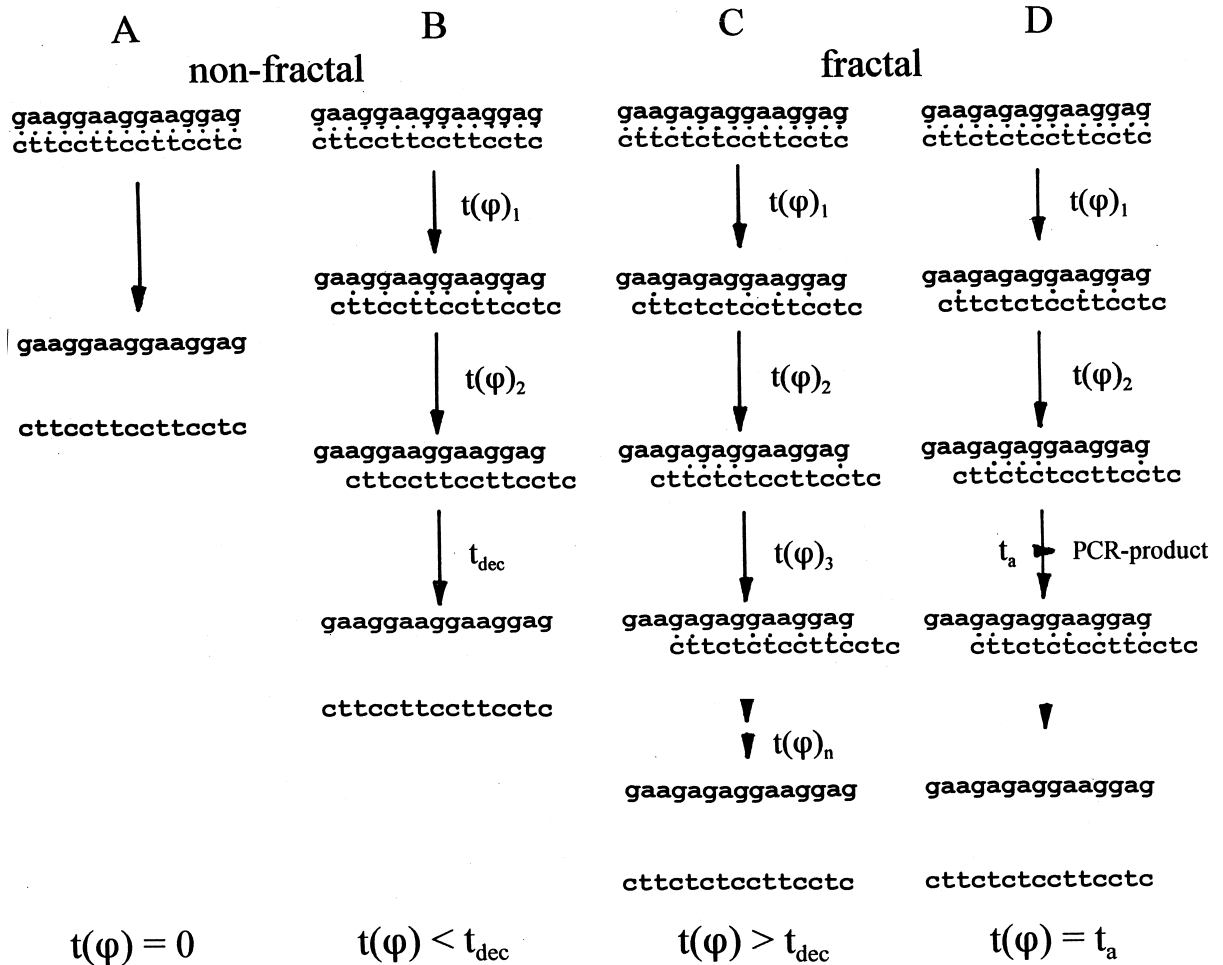


Fig. 2. Primer/template duplex formation and temporal evolution. (A) and (B), dissociation of the non-fractal primer occurs rapidly after shifting due to a short time for temporal evolution of the binding or duplex state. (C) and (D), the duplex formed with the fractal primer withstands rapid decoherence due to a sufficient number of binding states invariant to the extent of shifting. The binding state of a fractal sequence lasts longer than that of a non-fractal sequence. (D) The temporal evolution of the binding state is prematurely terminated due to collision with Taq polymerase fixing a particular state. This state can be binding and the DNA is amplified or anti-binding and the duplex dissociates.

F1s(forward):

5'gaagagaggaaggagttccatcatcaggatgCGGctagac3'

F1a(reverse):

5'gaagagaggaaggagttaggggctgaacctccacac3'

F2: 5'gaagagaggaaggag3'

R1s(forward):

5'gaaggaaggaaggagttccatcatcaggatgCGGctagac3'

R1a(reverse):

5'gaaggaaggaaggagttaggggctgaacctccacac3'

R2: 5'gaaggaaggaaggag3'

2.2. Methods

All PCR reagents were first prepared as master mixes following the instructions given by the manufacturer (Gibo BRL). Thermocycling was performed using a reaction volume of 25 μ l and the following amplification reaction: 2 min at 95°C; 35 cycles (30 s at 94°C, 30 s at various annealing temperatures (T_a), 30 s at 68°C); followed by final extension for 10 min at 72°C. As shown in Fig. 1,

the ST-II specific cDNA was first amplified with the primer combinations R1s/R1a or F1s/F1a at $T_a = 58^\circ\text{C}$, giving rise to an amplification product of 500 bp endowed with the fractal or non-fractal primer sequence at both ends of the DNA strand. The amplification product was then purified by means of agarose gel electrophoresis and an amount of 1 ng DNA was used as a template for the second round of PCR with the primer R2 or F2 (Fig. 1). The annealing temperatures were chosen as indicated in Fig. 3. The amount of amplification product was determined by ethidium bromide staining of the gel electrophoretically separated DNA followed by densitometric analysis. The concentration of Taq polymerase was determined by a Lowry protein assay following a modification by Wang and Smith (1975).

3. Results and discussion

3.1. Thermodynamic analysis of the primer/template duplex stability by PCR

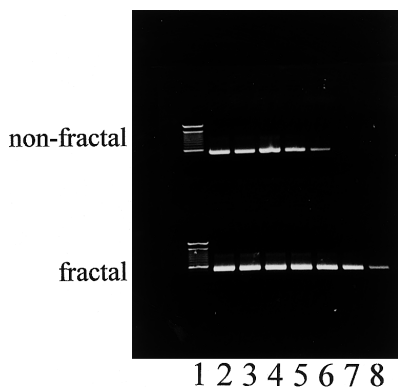


Fig. 3. PCR amplification in dependence on the annealing temperature. A 500 bp template generated as depicted in Fig. 1 was amplified with the non-fractal (top) or fractal (bottom) primer. The amplification product was separated by agarose gel electrophoresis and visualized by staining with ethidium bromide. A single amplification reaction was completed after 35 thermocycles. Each lane corresponds to an increment of the annealing temperature by 1°C starting with 52°C in lane 2 (lane 1, standard DNA). Each amplification reaction was repeated five times.

In order to determine the stability of the primer/template duplex, the PCR amplification was performed under identical conditions except that the annealing temperature was raised step-wise for each amplification reaction consisting of 35 thermocycles, as shown in Fig. 3. The amplification product was then separated by agarose gel electrophoresis and stained with ethidium bromide. Densitometric analysis of the staining intensity was used for product quantification. Reduction of the staining intensity by 50% indicated that the amplification reaction was performed at the melting temperature (T_m) of the primer/template duplex. In Fig. 3, it can be seen that T_m is 2°C higher for the fractal primer ($T_m(\text{F2}) = 58^\circ\text{C}$; Fig. 3, lane 8) than for the non-fractal primer ($T_m(\text{R2}) = 56^\circ\text{C}$; Fig. 3, lane 6). This result contrasted with the T_m calculated on the basis of the oligonucleotide sequence according to the method by Rychlik (1995), predicting $T_m(\text{R2}) = 39.9^\circ\text{C}$ and $T_m(\text{F2}) = 39.1^\circ\text{C}$. The Rychlik method is generally accepted for calculation of the primer/template duplex stability and predicts a higher melting temperature for an oligonucleotide sequence with identical purine residues as nearest neighbors (Petruska et al., 1988; Rychlik et al., 1990; Rychlik, 1995; Breslauer et al., 1996). As shown in Fig. 2, this was achieved in the non-fractal primer by permutation of the fifth and sixth residue. It should be noted that this permutation was the only difference between the sequence of the fractal (F2) and non-fractal (R2) primer. The observation that the melting temperature of the two different primer/template duplexes was about 20°C higher than calculated is common to short oligonucleotide sequences and due to the specific conditions of the PCR reaction. This, however, does not affect the prediction arising from the nearest-neighbor effect inherent to the non-fractal primer sequence.

Apparently, the amplification reaction was affected by additional factors, which are inherent to the primer sequence, but different from nearest-neighbor effects. It was assumed that these effects result from an attempted separation of primer and template due to a temperature-driven increase of kinetic energy. This separation can be initiated by shifting between template and primer upon destab-

bilization of the DNA duplex. Fig. 2 depicts the temporal evolution of a stable binding state $t(\varphi)$ as compared with a temperature-induced decoherence at time t_{dec} . Any stable binding state is assumed to collapse rapidly upon shifting for the non-fractal primer ($t(\varphi) < t_{\text{dec}}$) but may persist longer for the fractal primer ($t(\varphi) > t_{\text{dec}}$). This assumption is best explained by the observation that shifting of the fractal primer sequence along the template generates a more stable base pairing

than shifting of the non-fractal sequence. As shown in Fig. 4, this can be estimated by calculation of the binding energies contributed by base pairing upon shifting. The diagonals of the matrices represent a shift of the row (primer) sequence along the column (template) sequence by one base residue to the right (upper half) or left (lower half). A matching base pair containing three hydrogen bonds (c/g) is stabilized by the apparent binding energy of $\Delta G^* = 9 \text{ kJ/mol}$, and

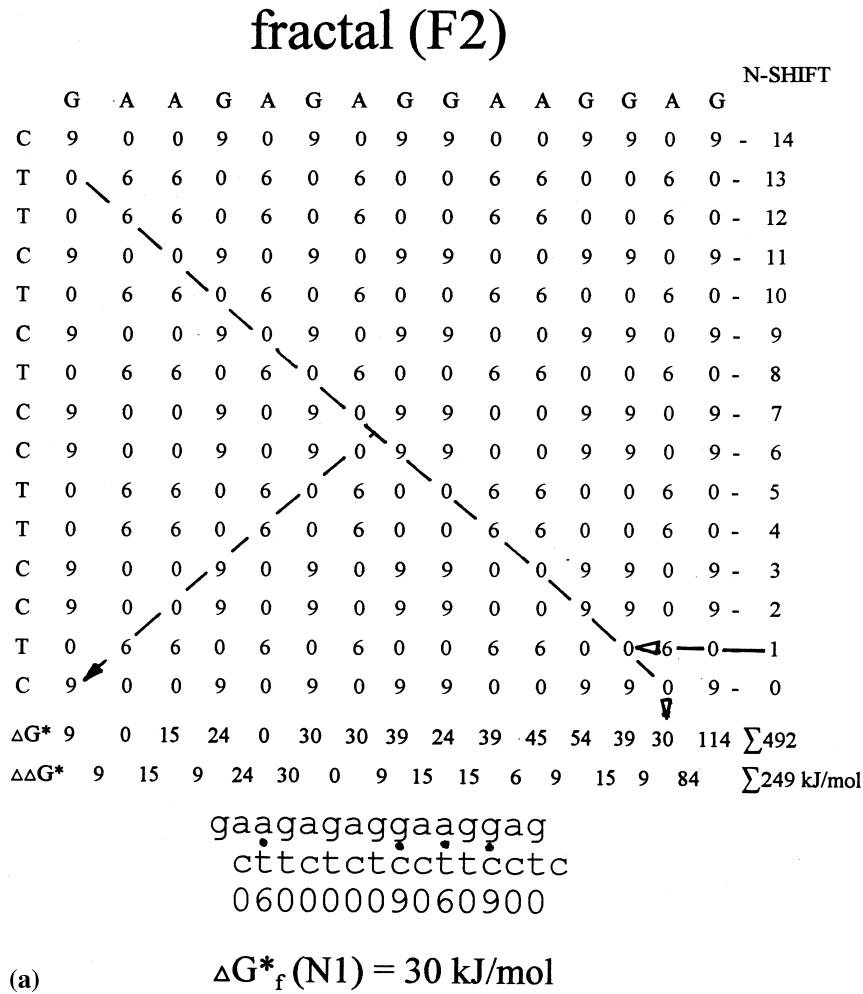


Fig. 4. Matrix notation for consecutive shifting of primer and template. The matrix shows binding energies upon consecutive shifting of primer (column) and template (row). The diagonal corresponding to a particular binding state can be found at the respective shift size N by drawing a horizontal line as shown. The binding energy is indicated at the perpendicular axis initiated from the crossing point of the horizontal with the diagonal axis. Shown is an example for $N = 1$. The diagonal with an arrow indicates the evolution of consecutive shifts.

non-fractal (R2)

	N-SHIFT															
	G	A	A	G	G	A	A	G	G	A	A	G	G	A	G	
C	9	0	0	9	9	0	0	9	9	0	0	9	9	0	9	14
T	0	6	6	0	0	6	6	0	0	6	6	0	0	6	0	13
T	0	6	6	0	0	6	6	0	0	6	6	0	0	6	0	12
C	9	0	0	9	9	0	0	9	9	0	0	9	9	0	9	11
C	9	0	0	9	9	0	0	9	9	0	0	9	9	0	9	10
T	0	6	6	0	0	6	6	0	0	6	6	0	0	6	0	9
T	0	6	6	0	0	6	6	0	0	6	6	0	0	6	0	8
C	9	0	0	9	9	0	0	9	9	0	0	9	9	0	9	7
C	9	0	0	9	9	0	0	9	9	0	0	9	9	0	9	6
T	0	6	6	0	0	6	6	0	0	6	6	0	0	6	0	5
T	0	6	6	0	0	6	6	0	0	6	6	0	0	6	0	4
C	9	0	0	9	9	0	0	9	9	0	0	9	9	0	9	3
C	9	0	0	9	9	0	0	9	9	0	0	9	9	0	9	2
T	0	6	6	0	0	6	6	0	0	6	6	0	0	6	0	1
C	9	0	0	9	9	0	0	9	9	0	0	9	9	0	9	0
ΔG^*	9	0	15	24	9	15	45	39	9	30	75	54	9	45	114	$\Sigma 492$
$\Delta\Delta G^*$	9	15	9	15	6	30	6	30	21	45	21	45	36	69		$\Sigma 357 \text{ kJ/mol}$

g	a	a	g	g	a	a	g	g	a	g	g	a	g	g	a	g
c	t	t	c	t	t	c	t	t	c	t	t	c	t	t	c	t
0	6	0	9	0	6	0	9	0	6	0	9	0	6	0	9	0

(b) $\Delta G^*_{\text{nf}}(\text{N1}) = 45 \text{ kJ/mol}$

Fig. 4. (Continued)

a matching base pair containing two hydrogen bonds (a/t) by $\Delta G^* = 6 \text{ kJ/mol}$ (Aboul-ela et al., 1985). These calculations were derived by a subtraction of the binding energies contributed by non-matching base pairs (a/g, a/c, t/g, t/c), which were then assigned to have a binding strength of 0 kJ/mol. It can be calculated that, upon shifting, the fractal and non-fractal primers have the same combined binding strength of $\Delta G^* = 492 \text{ kJ/mol}$ for matching base pairs, as indicated at the bottom of the matrices (Fig. 4). However, the difference between the binding strengths upon two shifts by one base residue is lower for the fractal

primer, suggesting that the probability for consecutive or reversible shifts is higher. This is indicated by the sum of the differences between the binding energies of two consecutive shifts, which was calculated to be 249 kJ/mol for the fractal and 357 kJ/mol for the non-fractal primer (Fig. 4). A thermodynamic analysis will take into account all permutations caused by random shifting of binding between two matching bases according to a Boltzmann distribution of binding energy. The partition function (Q) for the distribution of all accessible energy states (ϵ_j) of a system with N molecules is given by (Hill, 1986):

$$Q = \frac{1}{N!} q^N \quad \text{with } q = \sum e^{-\varepsilon_j/kT} \quad (1)$$

where k is the Boltzmann constant.

In a Bose–Einstein state, the independence of single states vanishes in favor of the sum of all binding energies (E_j):

$$Q = \sum e^{-E_j/kT} \quad (2)$$

It is assumed that a probability of shifting without losing matching base pairs facilitates a transition from classical Boltzmann to Bose–Einstein statistics. This would yield an additional binding potential (μ) arising from the superposition of potential shift operations resulting in a coherent binding state. In other words, the coherent state is characterized by a cooperative effect of binding states on the primer/template duplex formation. As discussed later, this may arise from electron tunneling between base pairs. By replacing E_j with $\varepsilon_j - \mu$ (less than ground state energy $\varepsilon_0 = 0$) in Eq. (2), the binding energy between primer and template is always higher in a Bose–Einstein state than with classical thermodynamics. The additional binding potential μ can be determined by comparing the free enthalpy of the primer/template duplex formation with the energy dispersion of a potential Bose–Einstein state.

The calculation of the free enthalpy for primer/template duplex formation with dependence on the template concentration (c_T) is derived from T_m according to (Aboul-ela et al., 1985):

$$\Delta G^\circ(T_m) = RT_m \ln(c_T/4) \quad (3)$$

The difference between the fractal and non-fractal primer binding strengths (ΔG^* in Fig. 4) may then account for the additional binding potential μ of the fractal primer due to the superposition of potential shifts. However, thermodynamic analysis cannot clearly demonstrate that this superposition originated in a quantum coherent state. One may argue that the shifting does occur, but the primer/template duplex stays partially unseparated between two shifts. A potential quantum coherence without adopting classical (single) binding states can only be evaluated by a kinetic analysis of the temporal evolution of the primer/template duplex formation.

3.2. Kinetic analysis of primer/template duplex formation

The temporal evolution of the primer/template duplex formation was analyzed by evaluating the Taq polymerase concentration dependent velocity of the amplification reaction. If the duplex formation proceeded without quantum coherent superposition of binding states, the concentration of the amplification product (c_A) increased over time in a linear correlation to the concentration of the duplex (c_D) and the enzyme (c_E) in a second-order reaction (Rychlik et al., 1990):

$$\frac{dc_A}{dt} = c_D c_E k_{II} \quad (4)$$

Fig. 5 shows the amount of amplification product obtained with the fractal and non-fractal primers with increasing concentrations of Taq polymerase as analyzed by agarose gel electrophoresis (Fig. 5A) and densitometric analysis (Fig. 5B) of the ethidium bromide stained DNA. The melting temperatures determined in Fig. 3 ($T_m(R2) = 56^\circ\text{C}$ and $T_m(F2) = 58^\circ\text{C}$) were chosen as the annealing temperatures in order to run the reaction at a state of 50% melting. The amplification reaction with the non-fractal primer revealed a linear dependence on the enzyme concentration, whereas the fractal primer showed only slightly more product with enzyme concentrations between 25 and 100 nM Taq polymerase (lanes 2–4). At a very low enzyme concentration (e.g. 10 nM), the amplification with the fractal primer was even more effective than with the non-fractal oligonucleotide (lane 2). At a higher concentration of Taq polymerase (e.g. > 100 nM), the two primers yielded almost the same amount of amplification product (lane 7). This distribution was unchanged upon alteration in the number of cycles (not shown). As shown in Fig. 5B, the amplification reaction with the fractal primer is not linearly dependent with the enzyme concentration, but sigmoidal with a shoulder at concentrations up to 1.0 μl enzyme (100 nM final concentration) added to the reaction mixture. This result can be explained by the assumption that an increasing amount of Taq polymerase lowered the concentration of primer/template duplex by a degree

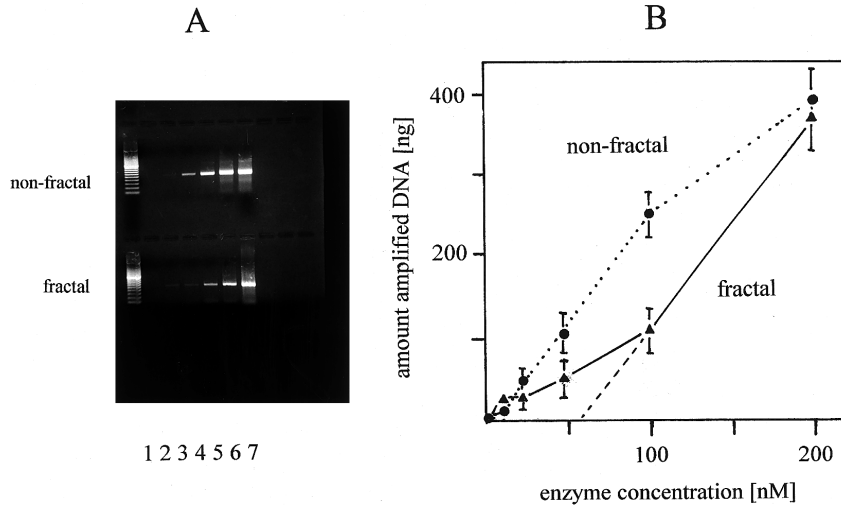


Fig. 5. PCR amplification in dependence on enzyme concentration. The PCR amplification was performed at the annealing temperature $T_a = 56^\circ\text{C}$ for the non-fractal (top) and $T_a = 58^\circ\text{C}$ for the fractal (bottom) primer with different concentrations of Taq polymerase. The amplification product was separated by agarose gel electrophoresis, visualized by staining with ethidium bromide (A), and the amount of DNA determined by UV-densitometric analysis (B). Lane 1, standard; lanes 2–7, 10, 25, 50, 100, 200 nM Taq polymerase. Each amplification reaction was repeated five times. Standard variations are indicated as bars.

compensating for the linear increase of the reaction velocity. As shown in Fig. 2D, this may have arisen from a premature dissociation of the primer/template duplex due to binding of the enzyme.

In the following analysis, it is assumed that the formation of the duplex with the fractal primer involves a temporal evolution of superimposed or quantum coherent binding states. In the case of primer/template duplex formation, this can be described by a superposition of rotational states due to helical winding of the two DNA strands around each other. Helical unwinding by 35° results in shifting of primer and template by $\Delta N = 1$. The time evolution ($U(\varphi)$) of the quantum coherent superposition of two rotational states with a separation energy of ΔE can be described by (Ioffe et al., 1999):

$$U(\varphi) = e^{-i\varphi(t)} \quad \text{with} \quad \varphi(t) = \Delta E t / \hbar \quad (5)$$

As shown in Fig. 2, this evolution would be disturbed by two different environmental effects resulting in decoherence, either driven by temperature or induced by binding to Taq polymerase. The first process is obvious since the uptake of kinetic energy by heating will separate the primer

and template DNA. The second process can be explained by the assumption that binding of the enzyme to the primer/template duplex collapses the superposition by fixing a particular binding state. This is comparable with a demolition measurement terminating a quantum coherent state. The thermally induced random shifting of primer and template DNA is no longer reversible, but fixed as a result of binding to the enzyme (Fig. 2D). Since the superposition involves binding and anti-binding states, it is very likely that the number of duplexes decohering to anti-binding states will not be amplified, but will dissociate after collision with the enzyme. The dissociation kinetics are then correlated to the concentration of enzyme due to a higher probability of enzyme binding with increasing enzyme concentration. The time interval between two collisions of the enzyme with the primer/template duplex will be lowered to an extent that it terminates the temporal evolution of the binding state ($\varphi(t)$) with $t(\varphi) = t_{\text{coll}} = t_{\text{dec}}$ (Fig. 2D). The dissociation kinetics of the primer/template duplex can be analyzed by first determining the amplification rate with the non-fractal primer, neglecting any short-lived quantum coherent states. A PCR amplification

during n cycles proceeds by duplication of a fraction $c_T k'_1 = c_D$ of the template concentration in each cycle according to:

$$c_A = c_D 2^n = c_T (1 + k'_1)^n \quad (6)$$

Note that the prime indicates a normalization of time onto the period of one amplification cycle. It follows from Eq. (5) that if $c_E \gg c_T$, the amplification reaction proceeds as a pseudo-first-order reaction with:

$$k'_1 = c_E k_{1I} \quad (7)$$

In Fig. 5B, it can be seen that at the shoulder of the plot using 25–100 nM Taq polymerase, with $c_T = 1$ ng/25 μ l, and $n = 35$, k'_1 is 0.14/cycle for the fractal and 0.17/cycle for the non-fractal primer. This can be explained by a faster dissociation of the fractal primer/template duplex induced by binding of Taq polymerase. According to the model already discussed, association with the enzyme entails decoherence of the primer/template superposition into single binding or anti-binding states. The statistics of the reaction converts instantaneously from Bose–Einstein to Boltzmann, resulting in an increase of the dissociation rate. The free enthalpy for dissociation can be calculated from the pseudo-first-order reaction, with (Stryer, 1995):

$$\Delta G(\text{dis}) = RT \ln(kT/hk_1) \quad (8)$$

where $R = 8.314$ J/K, k (Boltzmann's constant) = 1.38×10^{-23} J/K, and h (Planck's constant) = 6.63×10^{-34} J s.

The contribution of the binding state superposition in the fractal primer/template duplex towards a reduction of the dissociation energy will then be approximated by:

$$\mu_{\text{dec}} = RT \ln[k'_1(\text{fractal})/k'_1(\text{non-fractal})] \quad (9)$$

Using $T = 329$ K, Eq. (9) gives a free enthalpy of -531 J/mol for primer/template dissociation due to decoherence induced by binding of the enzyme.

From Eq. (3), it follows that the increase of T_m due to quantum coherence by binding state superposition within the primer/template duplex can be approximated by:

$$\mu_{\text{coh}} = R[T_m(\text{fractal}) - T_m(\text{non-fractal})] \ln(c_T/4) \quad (10)$$

Using $c_T = 1.2 \times 10^{-10}$ M as the initial template concentration (equals 1 ng DNA of 500 bp length in a reaction volume of 25 μ l) and $\Delta T_m = 2$ K, Eq. (10) yields a free enthalpy of -403 J/mol, attributed to the formation of a Bose–Einstein state.

Taking into consideration that these calculations are based on indirect measurement, the two values for free enthalpy are still sufficiently close to indicate an additional binding potential μ resulting from a quantum coherent process for fractal primer/template duplex formation. If we take 500 J/mol or 10^{-21} J/molecule as an approximate binding potential μ contributed by the Bose–Einstein state, the quantum process will be run in the range of the thermal energy (4×10^{-21} J/molecule) of the reaction.

3.3. Quantum mechanical analysis of the primer/template duplex formation

The thermodynamic and kinetic effects are suggested to arise from a superposition of binding states due to a potential well with $\mu_{\text{dec}} = 500$ J/mol or 10^{-21} J/molecule. This potential well emerges at the very low $\Delta\Delta G^*$ values or separation energies (Fig. 4) for step-wise primer/template shifts with the fractal primer sequence. As shown in Fig. 6, the width of the well is depends on the shift size $N = \omega_D t / 35^\circ$ (ω_D is the rotation angle for unwinding of the primer/template duplex in B-conformation of the DNA) with sufficiently high binding energy ΔG^* to withstand decoherence at ΔG_{cut} . The angle of 35° corresponds to the unwinding by one base pair ($\Delta N = 1$). The rotation and shifting motion is described by three vectors for the momentum direction (x, y, z) with x the axis for base pairing with ΔG^* , y the clockwise (+) or counter-clockwise (−) rotation by ω_D , and z the axis for shifting by N (Fig. 6). As discussed previously, since the time for collision with Taq polymerase (t_{coll}) is dependent on the enzyme concentration, there will be a threshold value for $t(\varphi) = t_{\text{coll}} = t_{\text{dec}}$ when the quantum superposition collapses due to demolition by collision with the enzyme (Fig. 2D). A

critical concentration is reached when the enzyme collides faster than the time needed for the undisturbed evolution of the superposition ($t_{\text{coll}} \leq t(\varphi)$). The Bose–Einstein state will then collapse to a probability distribution of binding and anti-binding states revealing Boltzmann statistics due to the distribution of kinetic energy in the heat bath (dotted graph in Fig. 6).

The decoherence time, t_{dec} , will be derived from the summation of the binding energies for base pairing after shift N , as given by the diagonals of the matrix shown in Fig. 4, e.g.

$$\begin{aligned} \text{diag}(N1)_{\text{fractal}} &= [06000009060900] = 30 \text{ kJ/mol} \\ &= \Delta G^*(N1) \end{aligned}$$

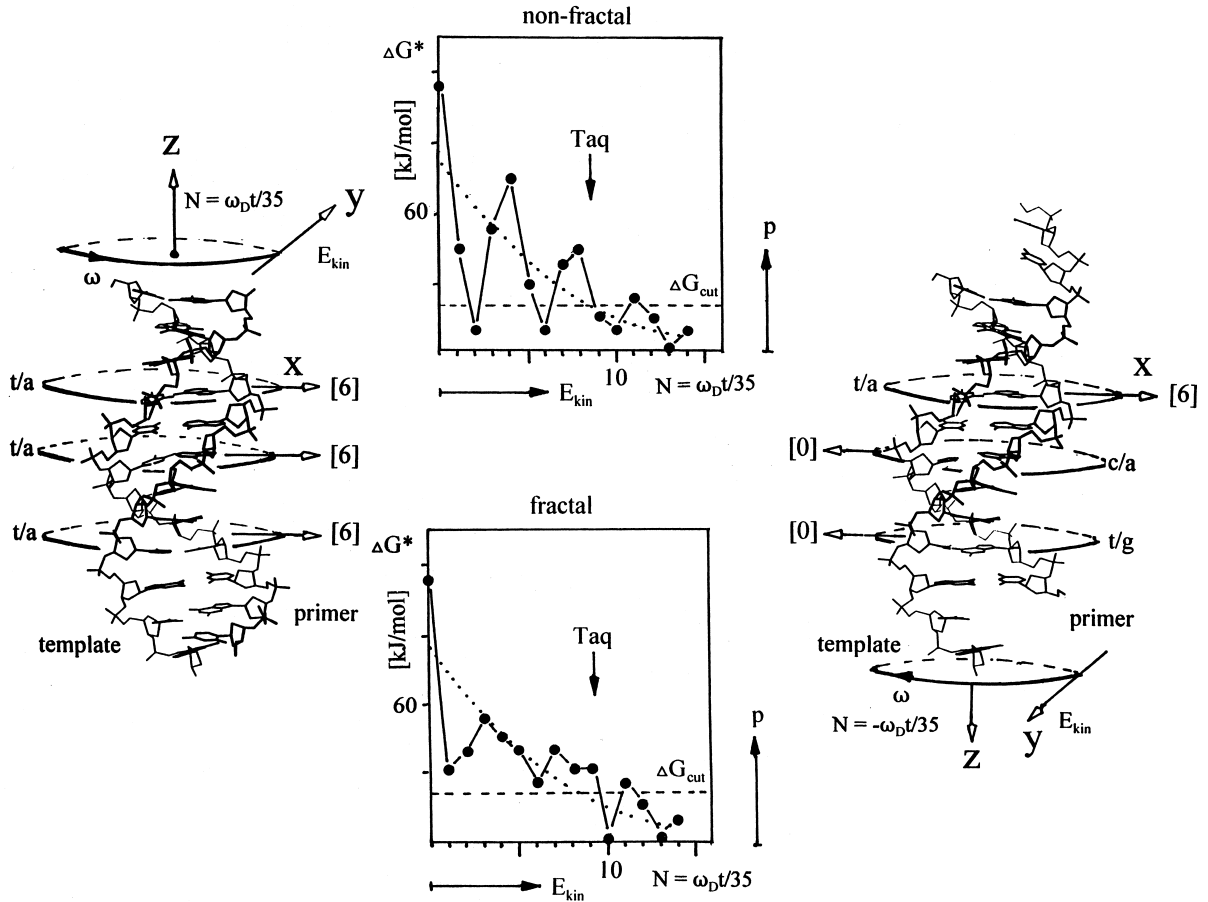


Fig. 6. Model mechanism for the primer template shifting. Shifting of the DNA by size N is induced by temperature-driven unwinding of the primer/template duplex by the angle $\omega_D t = N \cdot 35^\circ$ (35 corresponds to the angle upon rotation by one base pair or $\Delta N = 1$ on condition of a B-conformation of the duplex). The plots in the center show the summed binding energies ΔG^* for each base pairing upon shifting by N and are taken from Fig. 4. The arrow indicates the evolution of consecutive shifts. The probability (p) to find a particular duplex with the kinetic energy (E_{kin}) is an approximation and follows Maxwell–Boltzmann statistics (dotted graph). The duplex dissociates when E_{kin} is higher than ΔG_{cut} , the minimum binding energy for a stable duplex. The duplex can be stabilized by a superposition of binding states unless collision with Taq polymerase collapses the Bose–Einstein state. The coherent state is assumed to withstand dissociation in a potential well from $N = 1$ to $N = 10$ due to sufficiently low differences in binding energies upon shifting. The momentum directions for motions within the primer/template duplex are indicated: x , electron displacement due to hydrogen bonding; y , angular momentum for DNA unwinding; z , primer/template shifting upon unwinding and direction for electron tunneling.

$$\text{diag}(N2)_{\text{fractal}} = [0069690000009] = 39 \text{ kJ/mol}$$

$$= \Delta G^*(N2) \quad \text{etc.}$$

A different notation with the separation energies $\Delta\Delta G^*(i) = \Delta G^*(i+1) - \Delta G^*(i)$ between two consecutive shifts of size $\Delta N = 1$ (other shift sizes are neglected) is given by the following column matrix:

$$\mathbf{M}_{(\text{frac or non-frac})} =$$

$\Delta\Delta G^*(1)$	0	0	
0	$\Delta\Delta G^*(2)$	0	
0	0	$\Delta\Delta G^*(3)$	etc.

It is understood that this matrix is not complete since shift sizes of $\Delta N > 1$ are not included. The complete matrix would contain 2^n instead of n diagonal values. Furthermore, the off-diagonal values arising from the imaginary term in $\exp(-i\varphi)$ with ‘incomplete’ shifts of $\Delta N < 1$ (see Eq. (5)) are neglected in favor for a real-valued solution of the wave function $U(\varphi)$. These boundary conditions have been introduced in order to allow more intuitive access to the quantum mechanics of superimposed binding states by a numerical solution for $U(\varphi)$ derived from analysis of the matrix in Fig. 4. The arrow in Fig. 4 indicates the direction for the step-wise evolution of the separation energies $\Delta\Delta G^*$ between two consecutive shifts that appear as on-diagonal values in $\mathbf{M}$. The summed values or traces of the two matrices $\mathbf{M}(\text{frac})$ and $\mathbf{M}(\text{non-frac})$ are correlated with the binding potential contributed by a superposition of binding states within the fractal duplex (the non-fractal duplex is assumed to be non-coherent) according to:

$$\mu_{\text{dec}} = c \left[\sum \Delta\Delta G^*(i)_{\text{frac}} - \sum \Delta\Delta G^*(i)_{\text{non-frac}} \right] \quad (11)$$

For example, in Fig. 4, the difference of the diagonals yields a $\Delta\Delta G^*(N2, N1)$ value of 9 kJ/mol for the shift of the fractal and 36 kJ/mol for that of the non-fractal primer/template duplex. As shown in Fig. 6, this results in a binding energy $\Delta G^*(N2)$ of 39 kJ/mol for the fractal and 9 kJ/mol for the non-fractal primer. Since $\Delta G^*(N2)_{\text{non-frac}}$ falls below an approximated ΔG_{cut} for stable binding, we will assume that the non-fractal primer/

template duplex at $N = 2$ represents an anti-binding state. The coherent binding state is only possible if the separation energies $\Delta\Delta G^*$ are small. The summed values for $\Delta\Delta G^*(N1 \dots Nn)$ give a good approximation for the shift reversibility which is inversely correlated to the decoherence time or stability of the primer/template duplex. As shown in Fig. 4, the sums of the separation energies or traces of $\mathbf{M}$ are calculated to be 357 kJ/mol for the non-fractal and 249 kJ/mol for the fractal primer/template duplex. According to Eq. (11), the difference of the traces (108 kJ/mol) is then correlated to the binding potential μ of 500 J/mol for the escape from thermal decoherence due to superimposed binding states. The difference of the traces corresponds to the overall separation energy gap between the binding state $[1] = \text{primer/template duplex stays (binding)}$ or $[0] = \text{dissociates (anti-binding)}$. The constant (c) for calculation of μ has been introduced as empirical coefficient for linear correlation and can be approximated to 0.005. From Eq. (5), it can be concluded that the binding states $[1]$ or $[0]$ evolve by (μ expressed as 1×10^{-21} J/molecule):

$$U(\varphi) = \sum e^{-i\varphi(N)} \quad \text{with } \varphi(N) = \mu t_{\text{dec}}/h \quad (12)$$

The ground state φ_0 of the potential well is derived from a real-valued solution of Eq. (12) in a one-dimensional box, as shown in Fig. 7:

$$U(\varphi_0) = \sin v t_{\text{dec}} \quad (13)$$

with $1/t_{\text{dec}} = v$, the frequency of the wave function, equal to $\mu/h = 1.5 \times 10^{12}/\text{s}$.

The potential well extends over approximately $N = 10$ shifts consistent with DNA/primer unwinding by about $\omega_D = 350^\circ$ or a phase shift φ_0 ($[1] \rightarrow [0]$) for binding to anti-binding of $0.5 = 180^\circ = \pi$. From this shift, a decoherence time of $t_{\text{dec}} = 1/v = 0.7 \times 10^{-12}$ s can be calculated. The quantum mechanical interpretation of $U(\varphi_0)$, which corresponds to the amplitude of the sine function, is given by the square root of the probability of finding the primer/template duplex in a superposition within the window of $N = 10$. The value for $1/t_{\text{dec}}$ corresponds to that of the pseudo-first-order rate constant k_1 for dissociation induced by collapse of the quantum superposition.

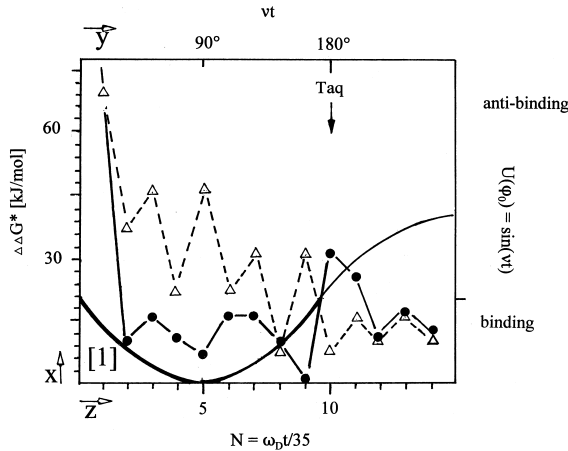


Fig. 7. Quantum superposition of binding and anti-binding states. The plot indicates the formation of a potential well for superposition of binding states due to low separation energies between consecutive shifts of primer and template DNA. The potential well is approximated by a sine wave function for superposition of a binding [1] and an anti-binding [0] state with a frequency ν of $10^{12}/s$. The arrow indicates the evolution of the binding state upon shifting. A phase shift of φ by $\pi/2 = 90^\circ$ corresponds to a DNA shift of $N = 5$ or helical unwinding of $\omega_D = 175^\circ$. A rotation of the DNA by $\omega_D = 350^\circ$ ($N = 10$) within $t_{dec} = 0.7 \times 10^{-12}$ s corresponds to a phase shift of the wavefunction by π (180°). The phase shift by π from binding to anti-binding can be induced by collision with Taq polymerase as indicated.

This is calculated from Eq. (8) with $\mu = 500$ J/mol for the potential contributed by quantum coherence:

$$k_I = \frac{kT}{e^{\mu/RT}h} = 6 \times 10^{12}/s \quad (14)$$

If the primer/template dissociation is induced by binding to Taq polymerase, then the concentration dependence can be approximated by Eq. (4) ($dc/dt = c_D c_E k_{II}$) with dc/dt the rate of DNA–enzyme complex formation, c_D the concentration of primer/template duplex $= 1.7 \times 10^{-11}$ M ($= c_T k_I$; see Eq. (6) with $n = 1$), c_E the enzyme concentration, and k_{II} the diffusion-controlled rate constant for DNA/enzyme complex formation, approximated on the basis of average collisions calculated for a variety of molecules in biological systems $\approx 10^9$ to 10^{10} (M/s) $^{-1}$ (Duke and Bray, 1999).

According to Fig. 4B, decoherence and dissociation resulting from anti-binding states induced by collision with the enzyme occurs at approximately $c_E = 10^{-7}$ M of Taq polymerase. The dissociation rate calculates to $dc/dt = 10^{-8}$ to 10^{-9} M/s, or in a reaction volume of 25 μ l, 10^{12} to 10^{13} molecule collisions/s. The average collision time of 10^{-13} to 10^{-12} s/molecule corresponds to the decoherence time calculated from Eq. (12). Thus, it can be consistently concluded that the decoherence of the coherent binding state is induced by collision with Taq polymerase. This explains why, in a certain concentration range, the fractal primer/template is less efficiently amplified than the non-fractal primer despite annealing at a higher melting temperature.

4. Conclusions

PCR amplification of DNA as an in vitro model for the detection of quantum processes in a biological system is useful for the following reasons: (i) the pairing of nucleotides provides a digital computation process of binding and anti-binding states that can be programmed by the sequence of the oligonucleotide primers; (ii) the amplification due to repetitive thermocycling allows for a very sensitive analysis of the binding reaction thermodynamics by determination of the efficacy of the product formation; (iii) the reaction kinetics can be directed by the concentration of enzyme; and (iv) the primer/template complementarity is a suitable model for molecular recognition that can be applied to other biologically significant molecular surfaces. In addition, the helical structure of the primer/template duplex allows for a translation of base-shifting into a rotation operation providing a convenient method for the computation of potential quantum processes.

The evolution $U(\varphi)$ of the primer/template duplex formation as given in Eq. (12) can be rewritten on the basis of a binding state superposition contributed by each base pairing after successive shifting steps (a and b coefficients for normalization of $[1]$ = binding state and $[0]$ = anti-binding state; $|a|^2 + |b|^2 = 1$):

$$U(\varphi) = 1/\sqrt{2}(a|0\rangle + ib|1\rangle) \quad (15)$$

In a classical system, there is only a transition from $|1\rangle$ to $|1\rangle$ or $|1\rangle$ to $|0\rangle$. A shift to the binding state $|1\rangle$ entails a stable DNA duplex, whereas $|0\rangle$ results in an irreversible dissociation of primer and template. In a quantum coherent system, however, the ‘bra-ket’ term $ib|1\rangle$ (with $b = \exp(-i\varphi(t))$) contributes to the additional binding potential arising from a reversible rotation of the two DNA strands around each other. The binding states are superimposed, ‘in between’ complementary base pairings in the x direction (see Fig. 6), and ‘compose’ the wave function $U(\varphi)$ in the z direction (see Figs. 6 and 7). The binding state vector rotates in the complex xy or Argand plane (Fig. 6) by $\varphi = vt$ (Fig. 7). Note that the rotation of the binding state vector arises from the rotation of the DNA but is not equivalent to it. In the Argand plane, the y component corresponds to the imaginary term $ib|1\rangle$, whose projection on the x axis yields the amplitude $U(\varphi) = \sin(vt)$ (real-valued solution, see Eq. (13)).

This quantum mechanical interpretation, of course, demands for the identification of a physical substrate that can actually undergo a binding state superposition. Since the DNA molecules are too heavy for a superposition upon shifting, we will assume in the following analysis that a binding state superposition results from an electron delocalization. Accordingly, the wave function $U(\varphi)$ will describe the actual binding orbital in the z direction corresponding to the axis of the double helix (Fig. 6). The amplitude of this function ($\sin(vt)$) is then equivalent to the square root of the probability for finding a ground state electron at a certain location within this orbital. The electron dispersion arises from sufficiently low separation energies between two consecutive binding states that allow for an electron tunneling in the z direction. The spatial limit for this tunneling is derived from the uncertainty relation $\Delta z \Delta p \geq \hbar$ (p is the momentum of the electron in the z axis, $\hbar$ is Planck’s constant). Assuming that the electron dispersion is driven by the thermal energy of the PCR reaction at 58°C, it can be calculated from the thermal deBroglie wavelength λ_e for a single electron that:

$$\Delta z = \lambda_e = \hbar/(2\pi m_e kT)^{1/2} \quad (16)$$

A single electron with the mass $m_e = 9.1 \times 10^{-31}$ kg will be dispersed by approximately 4 nm. This easily extends the length of a DNA stretch of ten stacked base pairs (3.5 nm) and is consistent with the potential well calculated from the wave function $U(\varphi)$. It should be noted that the classical electron orbitals for base pairing result from hydrogen bonding in the x direction (Fig. 6). In stable base pairings, these orbitals do not overlap due to Coulomb repulsion. The molecular orbital in the z direction is hypothetical and emerges from an electron excitation due to thermally induced shifting or rotation of the DNA strands. The kinetic energy conferred by the angular momentum in the y direction may be high enough to ‘loosen’ the hydrogen bonds in the x direction to a transition state and to excite electrons into a potential binding orbital in the z direction (Fig. 6). The energy band for this orbital or ‘ z -band’ should be above an energy gap of $c\Delta G_{\text{cut}}/6 \times 10^{23}$ (c is an empirical constant, estimated to be 0.005–0.025; see also Eq. (11)) as illustrated in Fig. 6. A fractal periodicity as designed within the primer sequence F2 is suggested to sustain the binding orbital in the z -band for 0.7×10^{-12} s, corresponding to an energy dispersion of 10^{-21} J/molecule ($= \Delta E = \hbar/t_{\text{dec}} \geq c\Delta G_{\text{cut}}/6 \times 10^{23}$). Accordingly, shielding of the Bose–Einstein state against thermally induced decoherence is a dynamic process counteracting a separation of primer and template by an ongoing binding potential arising from electron delocalization/tunneling. In the present study, the experimental approach by PCR amplification was sufficiently sensitive to detect this additional binding potential. It remains to be elucidated whether the tunneling effect can be determined by other experimental approaches, e.g. spectroscopic methods with a heated primer/template sample at a resonance frequency of 1.5×10^{12} Hz ($= \Delta E/\hbar$).

What is the biological significance of a binding state superposition? It is certainly not a critical contribution to overall binding in DNA or other molecules. A binding potential of 500 J/mol or 10^{-21} J/molecule is way too small to be crucial in an environmental heat bath of 4×10^{-21} J found

in a biological system. However, biological systems are based on mutual molecular recognition. This is relying on finding a matching binding state upon single molecular collisions. The collisions are the time-dependent steps rate-limited by their association kinetics. A binding state superposition due to quantum coherence, however, may stabilize a molecular complex for a sufficiently long time that fast sorting mechanisms can discriminate correct-from-false binding. Most recently, Patel has suggested that quantum coherent effects can be utilized for molecular recognition of the genetic code (Buchanan, 2000). He argues that the evolutionary optimization of using three bases for the coding of 20 amino acids emerges from a particular application of a quantum search algorithm introduced by Grover (1997). One out of $N = 20$ matching states can be found by superposition of $Q = 3$ binding states with $Q = \pi N^{1/2}/4$. This is suggested to accelerate a search from $N/2$ (classical) to $N^{1/2}$ (quantum) steps. It is assumed that the environment created by binding with an enzyme, in this case aminoacyl tRNA synthetase, maintains shielding against a heat bath for undisturbed quantum evolution of the superimposed binding states. Previously, this shielding effect was demonstrated for hydrogen tunneling in other enzymatic reactions (Ringe and Petsko, 1999).

In this study, however, molecular recognition of complementary surfaces takes place in an open system that is difficult to shield against decoherence effects induced by the environment. This is not only observed with binding of DNA strands, but also with molecular interactions of protein surfaces. In contrast to Patel's assumption, **interaction with the enzyme Taq polymerase induces decoherence of the superimposed binding states resulting in a fixed position of the primer/template duplex.** Only the fixed position allows for DNA amplification by enzyme-catalyzed chain elongation of the primer. However, the duplex stability prior to binding of Taq polymerase could be enhanced due to a potential quantum coherence of binding states. The generation of the quantum coherent superposition is suggested to arise from a specific geometry of binding states that can be generally described as self-similar or fractal distribution (Peitgen et al., 1992; Barnsley, 1993). Ide-

ally, the distance ratio between two binding states is the same at all scales. This leads to a scale-invariant probability for matching states regardless of the size for shifting of the molecules. The separation energy between two binding states may then be low enough to induce tunneling effects between electron orbitals as the result of thermally induced loosening of hydrogen bonds in the primer/template duplex. The thermal deBroglie wavelength of 4 nm is long enough to sustain an electron delocalization over several base pairs. This may be sufficient to overcome the separation energies for consecutive binding states in a fractal primer/template duplex but not in a non-fractal duplex.

The low additional binding energy of this binding state superposition, however, does not justify the implication of Grover's quantum search algorithm for single molecular recognition. As clearly demonstrated by the PCR reaction, the quantum effect was only detectable after exponential amplification. Therefore, Patel's suggestion to utilize a quantum search algorithm as a general mechanism for molecular recognition upon single molecule collisions is not realistic. Nevertheless, short-lived quantum coherence may play a role for sorting mechanisms that discriminate correct-from-false binding by fast recognition. Fast sorting has been described for proof-reading during DNA replication or protein translation at ribosomes (Stryer, 1995). Besides these specific proof-reading mechanisms, there are fast recognition events during protein or DNA folding that are common to all biological systems. It is still poorly understood how the search for protein folding can accomplish a fast correct-from-false sorting of one conformation within an average of 10^{35} possible structural states per protein. Intramolecular self-similar contact surfaces may stabilize domains that undergo short-lived quantum coherence of binding states during folding. Their evolutionary advantage may have resulted in a preference of fractal structures within the protein or DNA conformation. Recently, an analysis of 53 different proteins has shown that self-similarity in a scale range from 0.15 to 1.5 nm is characteristic for molecular contact surfaces (Goetze and Brickmann, 1992). It remains to be elucidated whether

this implies the implementation of quantum effects for molecular recognition. The model system presented in this paper may provide fundamental insight into the generation of quantum coherent binding state superposition in biological systems and may also be used for specific programming of the primer/template sequence for simulation of other binding state distributions.

Acknowledgements

The author wishes to thank Sarah MacKinnon for critically reading the manuscript. Part of the work was supported by an A.D. Williams grant.

References

- Aboul-ela, F., Koh, D., Tinoco, I., Jr., Martin, F.H., 1985. Base–base mismatches. Thermodynamics of double helix formation for dCA₃XA₃G + dCT₃YT₃G (X, Y = A, C, G, T). *Nucleic Acids Res.* 13, 4811–4824.
- Ahn, J., Weinacht, T.C., Bucksbaum, P.H., 2000. Information storage and retrieval through quantum phase. *Science* 287, 463–465.
- Barnsley, M.F., 1993. *Fractals Everywhere*. Academic Press, Cambridge, MA.
- Bieberich, E., Tencommen, T., Kapitonov, D., Yu, R.K., 2000. Effect of *N*-glycosylation on turnover and subcellular distribution of *N*-acetylgalactosaminyltransferase I and sialyltransferase II in neuroblastoma cells. *J. Neurochem.* 74, 2359–2364.
- Breslauer, K.J., Frank, R., Blöcker, H., Marky, L.A., 1996. Predicting DNA duplex stability from the base sequence. *Proc. Natl. Acad. Sci. USA* 83, 3746–3750.
- Buchanan, M., 2000. Life force. Do quantum computers make us what we are? *New Sci.* 2234, 20–24.
- Cha, Y., Murray, C.J., Klinman, J.P., 1989. Hydrogen tunneling in enzyme reactions. *Science* 243, 1325–1330.
- Duke, T.A.J., Bray, D., 1999. Heightened sensitivity of a lattice of membrane receptors. *Proc. Natl. Acad. Sci. USA* 96, 10104–10108.
- Goetze, T., Brickmann, J., 1992. Self similarity of protein surfaces. *Biophys. J.* 61, 109–118.
- Grover, L.K., 1997. Quantum computers can search arbitrary large databases by a single query. *Phys. Rev. Lett.* 79, 4709–4712.
- Hill, T.L., 1986. *An Introduction to Statistical Thermodynamics*. Dover, New York.
- Ioffe, L.B., Geshkenbein, V.B., Feigel'man, M.V., Fauchere, A.L., Blatter, G., 1999. Environmentally decoupled sds-wave Josephson junctions for quantum computing. *Nature* 398, 679–681.
- Kohen, A., Cannio, A., Cannio, R., Bartolucci, S., Klinman, J.P., 1999. Enzyme dynamics and hydrogen tunnelling in a thermophilic alcohol dehydrogenase. *Nature* 399, 496–499.
- Monroe, C., Meekhof, D.M., King, B.E., Wineland, D.J., 1996. A 'Schrödinger Cat' superposition state of an atom. *Science* 272, 1131–1136.
- Peitgen, H.-O., Jürgens, H., Saupe, D., 1992. *Chaos and Fractals*. New Frontiers of Science. Springer-Verlag, New York.
- Petruska, J., Goodman, M.F., Boosalis, M.S., Sowers, L.C., Cheong, C., Tinoco, I.J., 1988. Comparison between DNA melting thermodynamics and DNA polymerase fidelity. *Proc. Natl. Acad. Sci. USA* 85, 6252–6256.
- Ringe, D., Petsko, G.A., 1999. Tunnel Vision. *Nature* 399, 417–418.
- Rychlik, V., Spencer, W.J., Rhoads, R.E., 1990. Optimization of the annealing temperature for DNA amplification in vitro. *Nucleic Acids Res.* 18, 6409–6412.
- Rychlik, W., 1995. Selection of primers for polymerase chain reaction. *Mol. Biotechnol.* 3, 129–134.
- Seife, C., 2000. Cold numbers unmake the quantum mind. *Science* 287, 791.
- Stryer, L., 1995. *Biochemistry*. Freeman, New York.
- Tegmark, M., 2000. The importance of quantum decoherence in brain processes. *Phys. Rev. E* 61, 4194–4206.
- Wang, C.-S., Smith, R.L., 1975. Lowry determination of protein in the presence of Triton X-100. *Anal. Biochem.* 63, 414–417.